SATA P1/P13 physical orientation PROVISIONAL. P11 intentionally NC. No SATA 3.3 V use.

SATA 5V / USB diode OR -> TI 1A LDO. 12V only powers fans.

GPIO HIGH -> LVC inverter LOW -> NPN OFF -> fan PWM HIGH (100%).

External 10k PWM pullups default HIGH during reset. 5.23k TACH pullups -> 3V3 only.

CH340C 5V mode: V3 capacitor only. R3/R4 divide TXD: Q5/Q6 interlock auto BOOT/RESET.

NTC wiring: 3V3 -- fixed 10k -- ADC -- external 10k/B3950 NTC -- GND.

WIP: no manufacturing exports; connector continuity and current ratings require validation.

SATA P1 orientation and second locator unverified

Sheet: /
File: FanController.kicad_sch

Title: FanController P1 — ELECTRICAL PROTOTYPE / NOT FOR FABRICATION

Size: A2	Date:	Rev: P1-WIP
KiCad E.D.A. 10.0.7		Id: 1/1